

How Many Atoms, and Where: Predicting the Discrete Optima of Capacity, Minimax Estimation and the NPMLE Under Gaussian Noise

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Abstract—A transmitter limited in peak power, a statistician estimating a mean known to lie in a region, and a quantizer for a source spread over a region all choose a distribution for a bounded quantity seen through Gaussian noise. The optimum is discrete in all three; how many atoms it has, and where, is known only in part. We predict these optima from a smooth, possibly signed boundary density that would make the optimality condition an equality there, and from an edge layer of one-dimensional width. On a closed curve near a circle the optimal atoms are the nodes of the Szegő quadrature of that density, and the ring changes where a Verblunsky coefficient reaches ± 1 . Of twenty-four changes predicted in advance in the three problems, twenty fell within 0.61% of this rule, and ten within 10^{-4} with a first-order correction. The rule predicted 2×2 MIMO peak-power rings without atoms at the ends of the major axis; one, certified in ball arithmetic, contradicts a recent theorem. The density also predicted where a filled region's optimum first needs an inner atom. In up to five dimensions the edge constants fix optimal values on discs, balls, shells and annuli, through a widened volume (capacity), a perimeter law (NPMLE) or a widened Dirichlet eigenvalue (least favorable prior, strip up to 2.7% wider than in one dimension). A shift law fixed by $d = 2, 3$ predicted later center events of d -balls blind in $d = 4, 5$ but missed the first. Apart from the certificate, the results are numerical.

I. INTRODUCTION

Consider a transmitter limited in peak power, a statistician who knows that an unknown mean lies in a bounded region, and a quantizer for a source spread over a region. All three face the same mathematical situation: a bounded quantity is seen through additive Gaussian noise, and a probability law over that quantity has to be chosen optimally. For the transmitter the law is the input distribution that maximizes the mutual information. For the statistician it is the least favorable prior (LFP) of squared-error estimation. For the quantizer with squared-error distortion, the optimal reproduction law is found by fitting a Gaussian mixture to the source by maximum likelihood [1]; for a source spread uniformly, this fit is the population nonparametric maximum-likelihood estimate (NPMLE) of the mixing law. In all three problems the optimal law is discrete. Smith proved this for the scalar channel [2]. For the bounded normal mean it follows from Ghosh's work [3] (see [4] and [32, Prop. 4.18]), and Polyanskiy and Wu bounded the number of atoms of the one-dimensional NPMLE by a constant times the squared range of the data [5, Thm. 1]. On

a disc or a ball the capacity-achieving input sits on finitely many concentric circles or spheres [6], [7], [36].

How many atoms these laws have, and where the atoms sit, is known in much less detail. On an interval of half-length A the capacity-achieving input has between order $A\sqrt{\log A}$ and order A^2 atoms by the published bounds [8], [9], and about $A^{4/3}$ according to computations and a non-rigorous derivation [10], [11]. Every local change of the capacity-achieving support happens at the origin, where an atom appears or disappears, or the atom there splits into a pair or a pair merges [12]. In two dimensions even less is known. Dytso, Barletta and Kramer reduced the 2×2 multiple-input multiple-output (MIMO) channel with a peak-power constraint to an input confined to a filled ellipse [13]. They proved that for semi-axes up to $\sqrt{2}$ the optimal input lies on the boundary ellipse and is finite there. They also stated that both ends of the major axis always carry an atom, and they left open how the number of atoms grows as the ellipse becomes a circle.

We do not know of a description that predicts the number and the positions of the atoms of these optima on a region that is not a disc or a ball, or that predicts when an atom first appears inside the region. The three problems have also been treated one at a time, although their optimality conditions have the same form.

This paper proposes such a description and tests it on all three problems. It has two ingredients. The first is a smooth density on the boundary of the region, which we call the formal density: the law that would make the optimality condition hold with equality along the whole boundary, if a smooth law could do so. The formal density can be negative, so it is not itself an admissible law. Even so, the optimal atoms copy it. On a closed curve the optimal atoms sit at the nodes, with the weights, of the quadrature rule that reproduces the low Fourier coefficients of this density. A ring of K atoms has a Fourier coefficient of order K of its own, and the ring stops being optimal when the density's coefficient of order K reaches that value.

The second ingredient is an edge layer whose width comes from one dimension. Each problem has an edge constant on an interval, and on discs, balls, shells and annuli these constants fix the optimal value: for capacity and the LFP through the region widened by a one-dimensional strip, and for the NPMLE through a perimeter law.

Terms. A prediction is *registered* when its value and an

acceptance band were appended to a dated, append-only log before the optimum was computed or read, and *blind* when the optimum did not yet exist. A *ring* is an optimum whose atoms all lie on the boundary curve; it *copies* the formal density when its low Fourier coefficients equal the density's. A *loss event* is the parameter value at which an optimum stops satisfying the optimality condition; the ring is then *lost*. At a *birth* a new atom appears, at a *split* one atom becomes two, and at a *lift-off* the atom at the center of a ball becomes a small sphere. The *edge strip* of a problem is the widening of an interval described by its one-dimensional edge constant (Section III), and the *linear-response correction* is the first-order shift of a loss event derived in Section IV-C. Following each literature, we write A , m and R for the half-width or radius in the capacity, LFP and NPMLE problems; r_p and r_m are the semi-axes of an ellipse.

The results are the following.

- On closed curves near a circle, the optimal ring of atoms is the Szegő quadrature of the formal density, and it stops being optimal where a Verblunsky coefficient of that density reaches ± 1 . Of fourteen capacity events registered in advance, the ten on the two curves closest to a circle fell within 0.28% of the rule alone, and the six for which a linear-response correction was computed fell within 10^{-4} (relative). Ten more, in the NPMLE, the LFP and deterministic annealing, fell within 0.61% of the rule alone, four of them within 3.5×10^{-6} with the correction (Section IV).
- The rule predicted rings with no atom at the ends of the major axis. We certify one such ring in ball arithmetic, so the vertex statement of [13] does not hold (Theorem 1, Section V).
- The formal density predicts where the optimum on a filled region first needs an atom inside it. Registered predictions of that point held to 3×10^{-7} in all three problems, and for capacity the same construction predicted the mass of the new atom and the place where it splits (Section VI).
- On discs, balls, shells and annuli in up to five dimensions, the one-dimensional edge constants fix each problem's optimal value: a widened volume for capacity, with one constant fitted on discs; a perimeter law for the NPMLE; and a widened Dirichlet eigenvalue for the LFP. For the LFP the strip in $d \geq 2$ is wider than the one-dimensional one, by up to 2.7% at the sizes we could solve, and its limit for large regions is open (Section VII).
- In d dimensions the events at the center of a ball are displaced from their one-dimensional partners by an amount whose form is fixed by $d = 2$ and 3. It predicted the third to seventh events in $d = 4$ and 5 blind, missed the first two there by up to 13%, and failed at the first events in $d = 8$ (Section VIII).

Only Theorem 1 and Proposition 1 are proved here, the theorem with a computer-assisted certificate whose quadrature lemma is given in outline. Two further statements are proved only under a condition or in outline, and are labeled where they appear: the one-dimensional capacity expansion of Section III and the deformation identity behind the planar

NPMLE law of Section VII. The rest is numerical. The registration log, the solvers, the computed states and the logs of the runs reported here form the supplementary material, at <https://github.com/Tristan-tech-ai/how-many-atoms>; the time stamps of the log are not certified by a third party. Section II states the three problems and their common condition, Section III the one-dimensional facts, Sections IV–VI the planar results, Sections VII and VIII the results in d dimensions, and Section IX what does not carry over.

What the results are for, and what they do not show. Each of the three optima is used, and each use needs its atoms. A transmitter with a peak limit sends a finite constellation, and the capacity-achieving input is the constellation that carries the most information; the rule gives its points and their probabilities from a smooth density and a standard quadrature routine, without solving the optimization. A minimax estimator for a mean known to lie in a region is the Bayes rule against the least favorable prior, which otherwise has to be found by a search over discrete priors. Algorithms for the NPMLE work on a grid, and deterministic annealing follows splits as a temperature falls; the count and the predicted birth points say how fine the grid must be and where the splits come. The certified ring corrects a published statement: the optimal input of the 2×2 peak-limited MIMO channel need not use the ends of the major axis, so a design or a proof that assumes those atoms needs another argument. We have not compared these optima with practical schemes, so we cannot say how much rate, risk or likelihood a designer gains from them. Outside the certificate the results are numerical, and the closed-curve rule was tested near circles, on ellipses up to $r_p = 3$ and on one other family of curves.

II. THREE PROBLEMS AND ONE OPTIMALITY CONDITION

Let Z be a standard Gaussian vector in $\mathbb{R}^d$ with density φ_d , and let $\Omega \subset \mathbb{R}^d$ be a compact region. A probability law π on Ω is observed through the noise, $Y = X + Z$ with $X \sim \pi$, and its output density is $p_\pi = \pi * \varphi_d$. The three problems choose π as follows.

- *Capacity*: $C(\Omega) = \max_\pi I(X; Y)$, in nats.
- *LFP*: the Bayes risk of π for estimating X from Y under squared loss is $B(\pi) = \mathbb{E}|X - \mathbb{E}[X | Y]|^2$. The LFP maximizes B , and its Bayes risk is the minimax risk.
- *NPMLE*: for data with density f , the population NPMLE [37] maximizes $\int f \log p_\pi$ over π . For data uniform on Ω , with density $f_0 = 1/|\Omega|$ there, we report $G(\pi) = \int_\Omega \log(p_\pi/f_0) dy$.

Read with f as a source and squared error as the distortion, the NPMLE is the rate-distortion problem at the slope at which the test channel has unit variance [1]. In Section VI the source is Gaussian, $N(0, s^2 I)$ with $s = 2$, and the reproduction law is confined to Ω ; in Section VII the source is uniform on Ω .

Each objective is concave in π , so its maximum is characterized by its derivatives toward point masses. For a law π and

a point q , let

$$D_\pi(q) = D(\varphi_d(\cdot - q) \parallel p_\pi) \quad (\text{capacity}), \quad (1)$$

$$r_\pi(q) = \mathbb{E} |\delta_\pi(q + Z) - q|^2 \quad (\text{LFP}), \quad (2)$$

$$J_\pi(q) = \int \varphi_d(y - q) \frac{f(y)}{p_\pi(y)} dy \quad (\text{NPMLE}), \quad (3)$$

where $\delta_\pi(y) = \mathbb{E}_\pi[X \mid Y = y]$ is the Bayes rule. Moving a small mass from π to q changes the objective at the rate $D_\pi(q) - I(\pi)$, $r_\pi(q) - B(\pi)$ or $J_\pi(q) - 1$. Hence π is optimal if and only if the function is at most a constant on Ω , with equality on the support of π . The constant is C , $B(\pi)$ or 1 (for capacity see [2], [14] and Lemma 1 of [13]; for the LFP, [4] and [32, Ch. 4]; for the NPMLE, [38]). We call this function the optimality function of the problem and write it F_π when the problem does not matter.

Two identities are used later. By Tweedie's formula [33] and Stein's identity, the Bayes risk is $B(\pi) = d - I(p_\pi)$, where $I(p_\pi)$ is the Fisher information of the output density [15]; we checked this on the computed LFP of the disc of radius $m = 2.5$ to 2×10^{-12} . For capacity, the Hessian of D_π is $\nabla^2 D_\pi(q) = I - \mathbb{E}_Z \text{Cov}(X \mid q + Z)$ [16].

The atoms of an optimum are hard to predict one by one, but a smooth law is easier to compute. Let $\Gamma = \{z(t) : t \in [0, 2\pi]\}$ be a closed curve, and for a density g on Γ (with respect to $dt/2\pi$) let $F(t; g)$ be the optimality function at $z(t)$. The *formal density* of order M is the density $f(t) = 1 + \sum_{k=1}^M c_{2k} \cos 2kt$ whose optimality function has no harmonics of orders $2, \dots, 2M$, so that $F(\cdot; f)$ is constant along Γ to that order. We compute it by Newton's method in high-precision arithmetic. The optimality function of a signed f is defined wherever its output density is positive; for the truncations we use it was positive at every point where we evaluated it. Since the coefficients of f grow with k and f is negative on part of the curve from some order on, f is not an admissible law. We use only its low harmonics, which are what the optimal atoms follow.

III. ONE DIMENSION: EDGE CONSTANTS, COUNT AND CENTER EVENTS

On an interval $[-A, A]$ the optimal atoms are spaced unevenly in all three problems: they crowd near the two ends and thin out toward the middle. Each problem therefore has an edge layer, and a constant describes it (Table I). For capacity, $e^{C+h_1} = 2A + \beta$ up to exponentially small terms, where $h_d = \frac{d}{2} \log 2\pi e$ is the entropy of the noise; each end thus adds a strip of width $\beta/2 = 1.9908$ to the interval. This expansion is proved given a corollary that has not been audited separately (supplementary material): away from the two ends the output density equals e^{-C-h_1} up to a factor $1 + O(e^{-cA^{2/3}})$. Numerically the expansion agrees with this β to 5×10^{-12} on all 25 certified optimal states with A between 29.85 and 46.43. For the NPMLE, $G \rightarrow 2D$ as $A \rightarrow \infty$, a defect D per end. For the LFP, fits on $m \leq 40$ give the Bayes-risk deficit as $1 - B = \pi^2/(m + \delta)^2$ through order m^{-3} ; the leading term π^2/m^2 is Bickel's [17], and δ widens the interval at each end. Two response constants enter in Section VII: k , the response of the capacity strip to a linear reward on the

TABLE I
ONE-DIMENSIONAL EDGE CONSTANTS

constant	value	status
capacity strip β	3.98160794709	measured ^a
capacity reward response k	-0.63226	fitted ^b
NPMLE edge defect D	-0.304944111846	measured
NPMLE slope response $D'(0)$	$-0.041319 \pm 2 \times 10^{-6}$	post hoc ^c
LFP edge shift δ	3.34776	extrapolated ^d

^aOn 25 certified optimal states, to 5×10^{-12} . ^bAt $A = 16$ and 20.3; the $A = 16$ rows existed before registration; $A = 20.3$ replaced the registered $A = 20$, whose run was not accepted; a fit with k alone gives -0.62970 .

^cThe registered extraction gave -0.041438 ; the corrected value was confirmed blind at one width inside the fitted range. ^dFrom fits on $m \leq 40$; the registered check of the extrapolation at $m = 41, 43, 45$ is unscored.

input, and $D'(0)$, the derivative of the NPMLE defect when the data density has a small log-slope.

Numerically, the number of atoms grows like the $4/3$ power of the length, with the same constant in the three problems. With L the half-length in noise units, the count is $K \approx 0.4002 L^{4/3}$ plus terms of lower order. The constant $0.40024 = \frac{3}{2}(3/(16\pi^2))^{1/3}$ follows from a local spacing law: at depth x from an end the atoms have density ρ with $\rho^3 = 3x/(16\pi^2)$, and integrating ρ over the depth $0 \leq x \leq L$ from each end gives the count.

The exponent was found numerically by Mattingly et al. [10] and derived non-rigorously by Abbott and Machta [11], who print only the exponent. The constant and its agreement across the three problems come from our computations, which include forward tests, fixed before the computation, on twenty count changes for the LFP and for the NPMLE. Rigorous bounds on the count are the subject of separate work and are not used here.

As L grows, the support changes at the center of the interval: an atom is born at the origin, and later the atom at the origin splits into a pair, so births and splits alternate. For capacity every local change of the support takes place at the origin, by Corollary 2 of [12]; that births and splits alternate is what our computations show, for capacity as for the LFP and the NPMLE. The first events are classical. The binary input is optimal up to $A = 1.665925641$ [7]; Sharma and Shamai, who characterized this transition, reported 1.671 [18]. The two-point prior is least favorable up to $A = 1.0567$ [4]. The single atom of the NPMLE splits at $A = \sqrt{3}$, where the variance of the uniform data equals the noise variance; this is the first critical temperature of deterministic annealing [1]. Section VIII uses the whole sequence as the reference for births at the center of a ball.

IV. CLOSED CURVES: THE RING COPIES THE FORMAL DENSITY

A. Two curves and two problems

Let $d = 2$ and let the input be confined to a closed curve near a circle, or to the region the curve bounds. We use two families,

$$z(t) = Re^{it} + \eta(e^{-it} + \lambda e^{3it}), \quad R = r_p - \frac{\epsilon}{2}, \quad \eta = \frac{\epsilon}{2(1 + \lambda)}, \quad (4)$$

so that $z(0) = r_p$ and $\epsilon = r_p - r_m$. With $\lambda = 0$ the curve is the ellipse $(r_p \cos t, r_m \sin t)$; with $\lambda = 0.02$ it is a curve we call the mixed curve. Problem P1 confines the input to the filled region and P2 to the curve, and an optimum of P2 is an optimum of P1 exactly when its optimality function also stays below C inside. For the ellipse with $r_p \leq \sqrt{2}$ the two problems have the same optimum (Theorem 1 of [13]). Both curves are symmetric under $x_1 \mapsto -x_1$ and $x_2 \mapsto -x_2$, and so are their optima. A ring therefore consists of the pair X at the ends of the major axis ($t = 0, \pi$), the pair Y at the ends of the minor axis ($t = \pm\pi/2$), and orbits of four atoms at $t = \pm a, \pi \pm a$.

B. Para-orthogonal quadrature

The description borrows from the theory of orthogonal polynomials on the unit circle [19], [20]. Let $\alpha_0, \alpha_1, \dots$ be the Verblunsky coefficients of the measure $f(t) dt/2\pi$, where f is the formal density; they are defined by the Szegő recursion $\Phi_{n+1}(z) = z\Phi_n(z) - \alpha_n\Phi_n^*(z)$, with $\Phi_n^*(z) = z^n\overline{\Phi_n(1/\bar{z})}$ the reversed polynomial, as long as $|\alpha_j| < 1$. They are real by the symmetry, and $\alpha_j = 0$ for even j because f has only even harmonics. For $\tau \in \{+1, -1\}$, the K -node para-orthogonal rule replaces α_{K-1} by τ . If $|\alpha_j| < 1$ for $j \leq K-2$, the rule exists with K distinct nodes e^{it_k} on the unit circle and positive weights, and it reproduces the moments of orders $1, \dots, K-1$. The Verblunsky coefficients of a signed f need not lie in the unit disc; we use the rule only while they do. We place its nodes on the curve at $z(t_k)$. Because f has only even harmonics, the layout of the nodes is fixed by τ and K . For $\tau = +1$ there are nodes at $t = 0, \pi$ (the X pair), and at $t = \pm\pi/2$ (the Y pair) exactly when $K \equiv 0 \pmod{4}$. For $\tau = -1$ there is no X pair, and there is a Y pair exactly when $K \equiv 2 \pmod{4}$. For a ring with masses m_k at angles t_k , $\sum_k m_k = 1$, write $b_j = 2 \sum_k m_k \cos jt_k$ for its harmonics. The K -th moment of a measure depends on α_{K-1} only through the term $\alpha_{K-1} \prod_{j < K-1} (1 - \alpha_j^2)$, and the rule differs from the measure only in that coefficient. With the factor 2 that turns moments into cosine coefficients, this gives, for the rule's own harmonic of order K ,

$$c_K - b_K = 2(\alpha_{K-1} - \tau) \prod_{j < K-1} (1 - \alpha_j^2), \quad (5)$$

which we also checked to twelve digits on computed rules.

Conjecture 1 (copying). *While $|\alpha_j| < 1$ for $j \leq K-2$, the optimal K -atom ring is the K -node para-orthogonal rule of the formal density, up to copying errors $h_j = b_j - c_j$ ($j \leq K-2$) that tend to zero as $\epsilon \rightarrow 0$ at fixed j . It has $\tau = +1$ if it carries the X pair and $\tau = -1$ otherwise.*

Conjecture 2 (loss). *The K -atom ring stops being optimal near the ϵ at which $\alpha_{K-1} = \tau$, that is, where $b_K = c_K$. The kind of the event (an atom born or an atom split) and its place (the X or the Y pair) follow from the layout rule and the sign rule of Section IV-C.*

We state both at the precision of our computations, so that another group can falsify them. On the curves of Table III with $r_p \leq 1.2$, the loss event lay within 0.28% of the point

where $b_K = c_K$, and in the NPMLE, LFP and annealing events of Table II within 0.61%; on the ellipse $r_p = 3$ the miss was about 1% and fell with K . The linear-response form of Section IV-C is a sharper statement: where it was computed, the loss lay within 10^{-4} (relative) of its prediction, and it predicted the copying error h_{K-2} to within 2.2%.

Solved rings copy the density closely. On the ellipse with $r_p = 1.3$ and $K = 12$, the errors $|h_j|$ are at most 4×10^{-6} for $j \leq 8$, near the accuracy of the solver, and 6.2×10^{-4} at $j = 10$. The solved rings with $K = 10$ and 12 there have $h_{K-2} = -9.19 \times 10^{-4}$ and -6.21×10^{-4} , against -9.32×10^{-4} and -6.35×10^{-4} from the linear response of Section IV-C.

C. Why the ring copies, and the offset of the loss

The optimality function depends on the input only through the output density, and harmonic j of the input reaches the output density at radius ρ through the factor $I_j(\rho R)/I_0(\rho R) \approx (\rho R/2)^j/j!$. For a ring P near the density, therefore, $F(t; P) = F(t; f) + \sum_{ij} L_{ij} h_j \cos it$ up to second-order terms, with L nearly diagonal and $|L_{jj}|$ falling factorially in j . A stationary ring makes $g = F - C$ vanish with zero slope at each of its K atoms. Truncate g at harmonic K : a trigonometric polynomial of degree K that vanishes to second order at the K atoms is a multiple of $\Pi(t) = \prod_k (1 - \cos(t - t_k))$, so $g = \gamma \Pi$ for one number γ . Matching the harmonics below K then gives $h_j = 0$ for $j \leq K-2$ at leading order: $K/2 - 1$ equations for the $K/2 - 1$ free masses and angles, which is the para-orthogonal rule.

The loss condition $b_K = c_K$ tests a mean, while a ring is lost when a maximum of g reaches zero. Keeping the harmonics of g up to order $K+4$ and expanding them linearly about the formal density gives $g = \Pi q$, with q a trigonometric polynomial of degree 4; the linear-response event is the ϵ at which $\max_t q = 0$. Since $\max q \geq q_0$, the mean of q over t , and the copying condition is $q_0 = 0$ to leading order, the event comes before the copying event as ϵ decreases: the offset is positive, as all six linear-response cases of Table III show.

The sign of the second harmonic of q gives the place. It equals $-\text{sgn}(c_K c_{K+2})$, so when c_K and c_{K+2} have the same sign the maximum of q is at the Y pair, and otherwise at the X pair; an atom there splits, and a gap there gives a birth.

This argument is perturbative. We introduced the correction after the fact, on recorded events, before any blind test that uses it was registered.

D. The record

Each predicted value in Table III was appended to the log before any solve of the ring in question. The solves then tracked the optimum in ball arithmetic and bisected the first loss of the optimality condition. On the ellipse $r_p = 1$ the rule alone placed every event within 0.015%, and on the mixed curve within 0.28%. On the elongated ellipse $r_p = 3$ it missed by about one per cent, continuing a decreasing sequence of misses at smaller rings there. With the linear-response correction, computed for six events, the misses lie between 10^{-7} and 9.4×10^{-5} . Every registered kind and place was observed, except the kind of the 36-ring's loss. In the first

TABLE II
RING LOSSES REGISTERED IN ADVANCE IN THE OTHER TWO PROBLEMS, ELLIPSE $r_p = 1$, AGAINST THE TRACKED OPTIMUM. NPMLE: GAUSSIAN SOURCE, $s = 2$, ϵ INCREASING. DA: DETERMINISTIC ANNEALING AT $r_m = 0.7$, $s = 2$, TEMPERATURE DECREASING. ENTRIES ARE OBSERVED/PREDICTED -1 ; ALL REGISTERED BANDS WERE $\pm 1\%$ (RULE) AND $\pm 0.02\%$ (LR).

problem, K lost	kind, place	rule alone	LR
NPMLE, 14	Y birth	+0.217%	$\leq 3.5 \times 10^{-6}$
NPMLE, 16	X split	+0.019%	$\leq 3 \times 10^{-6}$
NPMLE, 18	X birth ^a	+0.61%	—
LFP, 14	Y birth	+0.076%	1.2×10^{-8}
LFP, 16	Y split	+0.057%	$\leq 5 \times 10^{-7}$
DA, 8	Y split	+0.37%	—
DA, 10	Y birth	+0.36%	—
DA, 12	Y split	+0.18%	—
DA, 14	Y birth	+0.31%	—
DA, 16	Y split	+0.007 to +0.039%	—

^aObserved; the kind of this event was not registered.

invalid state the optimality function is positive in the gaps next to the empty Y position and largest at the vertex. That fits the predicted birth, but it does not separate the birth from a split of the neighboring atoms. The rings of 32 and 34 atoms on the ellipse $r_p = 1$ carry no X pair, and Section V certifies one of them. The two valid 36-ring states we solved, at $\epsilon = 0.0310$ and 0.0308 in 448-bit arithmetic, carry none either: their atoms nearest the end of the major axis sit at $t = \pm 4.97^\circ$ and $\pm 4.96^\circ$, and $D_\pi - C$ at the end is -2.6×10^{-56} and -3.8×10^{-57} .

The same rule was tested in advance in the other two problems, on the ellipse $r_p = 1$ (Table II). Five ring losses of the NPMLE with a Gaussian source ($s = 2$) and of the LFP lay within 0.61% of the rule alone, and the four for which the linear-response correction was registered lay within 3.5×10^{-6} (relative) of it. Deterministic annealing gives a second path through the NPMLE. At fixed shape ($r_p = 1$, $r_m = 0.7$, $s = 2$) the temperature T is lowered, and the problem at temperature T is the NPMLE with noise of standard deviation $\sqrt{T/2}$ [1]. Its first five ring losses on cooling lay within 0.53% of the rule. Every registered kind and place in these ten events was observed. With Table III, twenty-four ring losses predicted in advance in the three problems fell inside their bands.

E. The count as the curve becomes a circle

Read as a count, Conjecture 2 says that the optimum has $K = j^* + 1$ atoms, where j^* is the smallest odd index with $|\alpha_{j^*}| \geq 1$. We evaluate it in the $K/2$ truncation, in which the formal density used for a K -atom ring has $K/2$ harmonic pairs. At $r_p = 1$ the loss points of the rule for $K = 16, 20, 24, 28$ and 30 are $\epsilon = 0.07171, 0.05835, 0.04902, 0.04266$ and 0.04080 , so $K\epsilon = 1.15, 1.17, 1.18, 1.19, 1.22$: in a first lobe the count grows like $1/\epsilon$ (Fig. 1). To leading order in ϵ the coefficients are $c_{2k} = 2a_k(\epsilon/2)^k$, and a_k first changes sign at $k = 17$, which ends the first lobe. Past it the rule gives $K\epsilon^{1/2} = 6.29, 6.31, 6.37, 6.45, 6.53$ for $K = 34, 36, 38, 40, 42$, a local exponent $d \ln K / d \ln(1/\epsilon) = 0.61$; the 34- and 36-rings were solved, the larger ones were not. From $K = 38$ to 320 the rule's exponent is 0.59. If Conjecture 2 holds, the layout of the optimal ring alternates in blocks about 30 wide

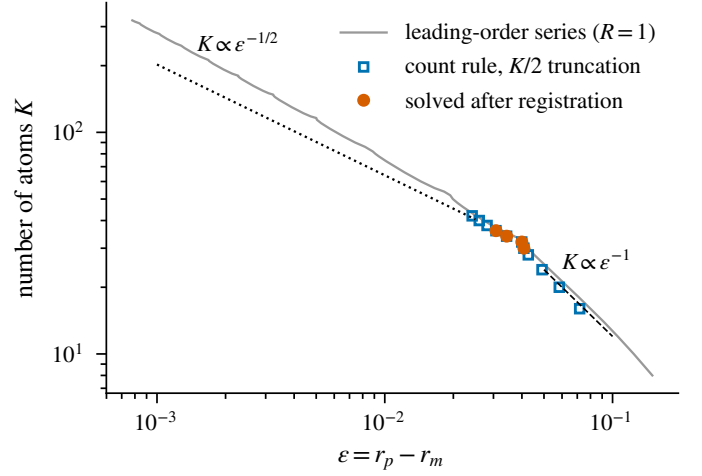


Fig. 1. The number of atoms grows like $1/\epsilon$ in a first lobe and more slowly beyond it. Ellipse $r_p = 1$: the count rule in the $K/2$ truncation at $R = r_p - \epsilon/2$ (squares), the leading-order series at $R = 1$ (line), optima solved after their prediction was registered (dots; rows 1–4 of Table III), and the guides $K = 1.2/\epsilon$ (dashed) and $K = 6.4\epsilon^{-1/2}$ (dotted).

in K (16 ring sizes each), so the vertex statement of [13] fails on infinitely many intervals of ϵ accumulating at zero. This extrapolates the sign pattern of a_k beyond $k = 160$ and is not a measurement.

F. Toward the segment

As $r_m \rightarrow 0$ the ellipse collapses onto the segment $[-r_p, r_p]$, and both problems become the scalar channel with $A = r_p$; a scalar input with K_1 atoms is the ring with $2K_1 - 2$ atoms. Evaluated at $r_m = 0.005$ and extrapolated linearly in r_m , the loss rule with the linear-response correction places the first two scalar changes at $A = 1.6653$ and 2.9317 , against 1.665925641 and 2.9075 (-0.04% and $+0.83\%$) [7], [12]; the copying condition alone gives 1.592 and about 3.2 . The procedure was fixed before it was run, but the targets were known. From the third change on ($A \approx 4.0$) the formal density near the segment violates $|c_n| \leq 2$, which every positive measure satisfies, and the rule is not defined there; the scalar count then follows the edge layers and the count law of Section III.

V. A CERTIFIED RING WITHOUT VERTEX ATOMS

A. The result

Theorem 4 of [13] states that the two ends of the major axis, $(\pm r_p, 0)$, belong to the support of the capacity-achieving input for every $0 \leq r_m \leq r_p$. The copying rule predicted rings without these atoms on the ellipse $r_p = 1$, for ϵ between 0.03077 and 0.0408049 , where the optimal rings have 32, 34 and 36 atoms. We certify one of them.

Theorem 1 (computer-assisted). *For $r_p = 1$ and $r_m = 0.9597$, the capacity-achieving input of the channel $Y = X + Z$ with X in the filled ellipse is a symmetric ring of 32 atoms whose angles and masses lie within 6.82×10^{-66} of the values in Table IV. In particular $(\pm r_p, 0)$ is not in its*

TABLE III

REGISTERED PREDICTIONS OF RING LOSSES AGAINST THE TRACKED OPTIMUM (CAPACITY). P1: FILLED REGION; P2: INPUT ON THE CURVE. FORM S: THE LOSS CONDITION $b_K = c_K$ OF CONJECTURE 2; LR: WITH THE LINEAR-RESPONSE CORRECTION. KIND AND PLACE: X OR Y PAIR, SPLIT OR BIRTH, REGISTERED WITH THE VALUE AND OBSERVED. EACH VALUE WAS REGISTERED WITH AN ACCEPTANCE BAND, AND ALL FOURTEEN OBSERVED EVENTS FELL INSIDE THEIR BANDS (BANDS IN THE SUPPLEMENTARY LOG).

curve, r_p , problem, event	form: registered ϵ	observed ϵ	observed/predicted -1	kind, place
ellipse, 1, P1, 30-ring lost	S: 0.0408032574	(0.0408049, 0.0408055)	+0.004 to +0.006%	X split
ellipse, 1, P1, 32-ring lost	S: 0.0399404809	(0.0399457, 0.0399463)	+0.013 to +0.015%	Y birth
ellipse, 1, P1, 34-ring lost	S: 0.0342692971	(0.0342733, 0.0342734)	+0.012%	Y split
ellipse, 1, P1, 36-ring lost	S: 0.0307639408	(0.0307656, 0.0307672)	+0.005 to +0.011%	Y birth (not resolved)
mixed, 1.2, P1, 16-ring lost	S: 0.134287189	0.13432752	+0.030%; LR 1×10^{-7}	X split
mixed, 1.2, P1, 18-ring lost	S: 0.12493239	0.1252793873	+0.278%	X birth
mixed, 1.2, P1, 20-ring lost	S: 0.100908	0.1010178	+0.11%; LR $\leq 7 \times 10^{-6}$	X split
mixed, 1.2, P1, 22-ring lost	S: 0.087130	(0.0872266, 0.0872344)	+0.11 to +0.12%	X birth
mixed, 1.2, P1, 24-ring lost	S: 0.074900	(0.07495, 0.07500)	+0.07 to +0.13%	X split
mixed, 1.2, P1, 26-ring lost	S: 0.065283	(0.0653262, 0.0653320)	+0.07 to +0.08%	X birth
ellipse, 3, P1, 18-ring lost	LR: 0.9518201483	0.9519100801	+0.0094%	Y birth
ellipse, 3, P2, 20-ring lost	LR: 0.9046658875	0.9047303949	+0.0071%	Y split
ellipse, 3, P2, 22-ring lost	LR: 0.8625141199	0.8625613848	+0.0055%	Y birth
ellipse, 3, P2, 24-ring lost	LR: 0.8242977666	0.8243330	+0.0043%	Y split

On the ellipse $r_p = 3$ the form S alone missed the 18-ring by +1.18%, after misses of +8.1, +4.8, +3.3, +2.3, +1.8 and +1.4% at $K = 6, \dots, 16$. The mixed curve at $r_p = 1.2$ was checked to satisfy P1 on an interior grid.

TABLE IV

THE CERTIFIED 32-ATOM RING AT $r_p = 1$, $r_m = 0.9597$: ONE REPRESENTATIVE PER ORBIT, MASS PER ATOM. MIDPOINTS OF THE CERTIFIED BOX ROUNDED TO 20 DECIMALS; THE BOX RADIUS IS AT MOST 6.82×10^{-66} .

orbit	parameter angle t_j (rad)	mass per atom
G1	0.04514041824277957604	0.02587729920137166890
G2	0.23892408883896762725	0.03897966931781938891
G3	0.45323176005362528202	0.03545268828046724276
G4	0.66308206341144781172	0.03249339856300210627
G5	0.86870324583193416037	0.03056684647659560933
G6	1.07128670703876574422	0.02940210562835088532
G7	1.27186053563821704426	0.02875615919397271245
G8	1.47128043156772716194	0.02847183333842038607

Atoms at $(r_p \cos t, r_m \sin t)$, $t = \pm t_j, \pi \pm t_j$.

support. The capacity is $C = 0.39064923014816081039 \dots$ nats (0.563588 bits), enclosed to 7×10^{-74} .

The capacity-achieving input is unique, since the output law determines the input law (the Gaussian characteristic function has no zeros) and the output entropy is strictly concave in the output law. The channel $Y = HX + Z$ with $H = \text{diag}(1, 0.9597)$ and $\|X\| \leq 1$ realizes this input. Theorems 1–3 of [13] are not affected, since their proofs do not use Theorem 4. The two atoms nearest the end $(r_p, 0)$ sit at $t = \pm 2.5864^\circ$ (Fig. 2). At the end itself the certificate gives $D_\pi - C = -1.1276 \times 10^{-50}$, and at the end of the minor axis -1.4177×10^{-48} , so neither is a support point.

B. The certificate

All arithmetic is midpoint-radius ball arithmetic (python-flint 0.9.0, the interface to FLINT and Arb [21]). The unknowns are the eight angles, the eight masses and C ; the equations are $D_\pi = C$ and $\partial_t D_\pi = 0$ at one atom of each orbit, and $4 \sum_j m_j = 1$. The proof has four steps, with every constant in Appendix A.

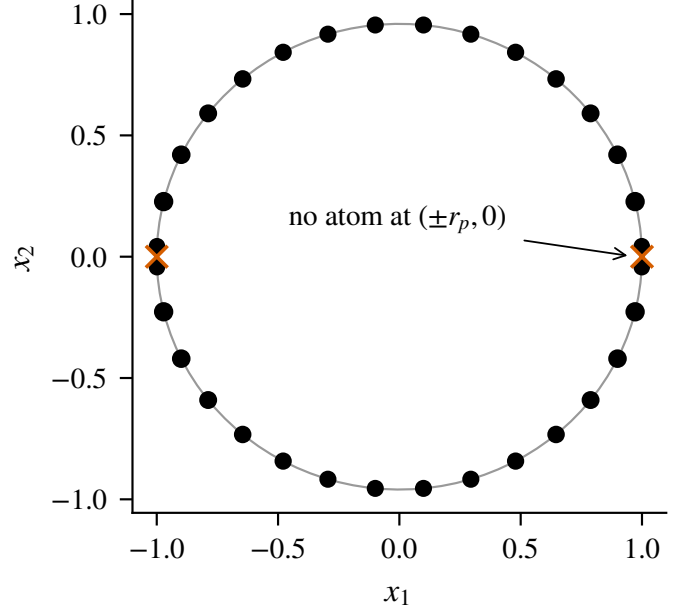


Fig. 2. The certified optimum carries no atom at the ends of the major axis. Ellipse $r_p = 1$, $r_m = 0.9597$; marker area is proportional to mass. The pair next to each end, at $t = \pm 2.5864^\circ$, carries the smallest mass, 0.0259.

- 1) *Quadrature with an error bound.* The expectation in D_π is replaced by a trapezoid sum on a square grid, with an error bound from analyticity in a strip [22] (Lemma 1).
- 2) *An exact solution.* A Krawczyk test [23]–[25] shows that a box of radius 6.82×10^{-66} around the computed ring contains exactly one solution of the optimality equations, with every mass above 0.025877; the capacity C , one of the unknowns, carries its own smaller radius in the same box.
- 3) *The boundary.* On 230 pieces of $t \in [0, \pi/2]$, Taylor enclosures with Cauchy remainders show $D_\pi - C \leq -1.74 \times 10^{-53}$ away from the atoms and $\partial_t^2(D_\pi - C) < 0$ in a zone around each atom, uniformly over the box.

- 4) *The interior.* The largest atom radius is $0.99992 < 1$, so by the Hessian identity of Section II D_π is strictly convex on $\mathbb{R}^2$, and every interior point lies strictly below the boundary maximum.

Together with the sufficiency of the optimality condition [2], [14], these steps prove Theorem 1, with one caveat: the proof of Lemma 1, which bounds the quadrature error, is given here in outline. A separate evaluator with no shared code (a trapezoid rule at 256 bits) reproduces $D_\pi - C$ below 10^{-76} at the atoms and the two vertex values to the digits shown. A second rigorous check shares nothing with the ring. Putting mass $f(t_i)/128$ on 128 equally spaced points of the curve, with f the formal density truncated at four harmonic pairs ($\min f = 0.899$), gives an admissible input whose mutual information bounds C from below, and its output law q bounds it from above through $C \leq \sup_x D(P_{Y|x}||q)$ [26]. With the same machinery this gives $0.3906492301481604 \leq C \leq 0.3906492301483361$ (ends rounded outward), which contains the certified value.

C. Where the published proof breaks

The proof of Theorem 4 in [13] lets $t < r_p$ be the largest abscissa in the support and uses their Lemma 7, $\partial D/\partial x_1 \geq x_1 - t > 0$ for $x_1 > t$, to conclude that $D(x_1, x_2) > D(t, x_2)$ for $x_1 > t$, which “would violate the KKT condition”. The implicit step is that some (t, x_2) with feasible points to its right is a support point. On the certified ring the largest abscissa is $t = \cos(2.5864^\circ) = 0.998981$, attained by the pair at $x_2 = \pm 0.043307$. On those two lines the ellipse ends at $x_1 = t$, so no feasible point lies to the right of the support points, and the comparison yields no violation. On the line $x_2 = 0$, where feasible points with $x_1 \in (t, r_p]$ exist, $(t, 0)$ is not a support point. What the argument does prove is the following.

Proposition 1. *Let $0 < r_m \leq r_p$, let t be the largest abscissa of a support point, and suppose $0 < t < r_p$. Then no feasible point lies to the right of any support point with abscissa t ; that is, every such point (t, x_2) satisfies $t = r_p(1 - x_2^2/r_m^2)^{1/2}$.*

Proof. At a support point $D(t, x_2) = C$. By Lemma 7 of [13], $D(x_1, x_2) > C$ for $x_1 > t$, which the optimality condition forbids at feasible points. $\square$

So the end $(r_p, 0)$ belongs to the support only if a support point of maximal abscissa lies on the axis. For $r_p \leq \sqrt{2}$, where every support point lies on the boundary, the argument gives nothing more.

VI. LEAVING THE BOUNDARY: BIRTHS INSIDE A FILLED REGION

A. The center test

On a filled region the boundary ring stays optimal while its optimality function also stays below the constant inside. On the grids we checked, the largest interior value sat at the center whenever it came close to the constant, so the boundary optimum is lost to an atom born at the center, the counterpart of the birth at the origin on an interval. The formal density predicts when this happens. Harmonic j of the input

TABLE V
REGISTERED CENTER BIRTHS, ALL PASSED: THE r_m AT WHICH THE CENTER EXCESS OF THE BOUNDARY RING VANISHES. MIXED λ : THE CURVE (4) WITH THAT λ . NPMLE: GAUSSIAN SOURCE, $s = 2$; b RINGS AT 200 GAUSS–HERMITE NODES (AT 120 NODES THE ELLIPSE VALUE IS 1.741648, INSIDE THE BAND). THE LFP ON THE MIXED CURVE USES $\lambda = 0.1$, WHERE ITS FORMAL DENSITY CONVERGES.

problem	region, r_p	registered (band)	observed
capacity	ellipse, 2.8	2.1874219 ($\pm 10^{-5}$)	2.18742185
LFP	ellipse, 1.7	1.4038104 ($\pm 2 \times 10^{-4}$)	1.4038104
NPMLE	ellipse, 2.1	1.7416352 ($\pm 2 \times 10^{-4}$)	1.741635 ^b
capacity	mixed 0.2, 2.8	2.1756059 ($\pm 2 \times 10^{-4}$)	2.175606
LFP	mixed 0.1, 1.7	1.4015695 ($\pm 2 \times 10^{-4}$)	1.4015695
NPMLE	mixed 0.2, 2.1	1.7363460 ($\pm 2 \times 10^{-4}$)	1.7363463 ^b

is suppressed near the origin by the factor of Section IV-C, so the value $F(0)$ depends mainly on the low harmonics, and those are the ones the ring copies. We therefore evaluate the optimality function of the formal density at the center, $F_f(0)$, and subtract C , taken as the mean of F_f over the curve; the difference is the center excess. On a disc $f = 1$ exactly, and the excess vanishes at radius 2.4534 for capacity (2.454 in Table I of [7]) and at 1.53499 for the LFP (1.535 in the same table). In that table the LFP value is the largest radius at which the uniform law on the circle maximizes the minimum mean-square error. For the NPMLE with a Gaussian source of standard deviation $s = 2$ the excess vanishes at 1.8982.

B. Registered center births

Table V lists the registered births and Fig. 3 the curves they lie on. The boundary rings were solved afterwards in 128- to 160-bit arithmetic, and their center excess was read at several values of r_m ; its zero is the observed birth. The number of atoms in the ring did not need to be resolved. The center excess was the same for LFP rings of 16, 18 and 20 atoms to 10^{-13} on the ellipse (18 and 20 atoms to 1.4×10^{-10} on the mixed curve), and for NPMLE rings of 16 and 20 atoms to 1.2×10^{-11} . Below about 14 atoms it still depends on the ring size. All six registered births passed, within 3×10^{-7} in r_m . Since the ring copies the low harmonics of the density, this tests a consequence of copying for an interior quantity; it is not independent evidence for Conjecture 1.

For the 2×2 MIMO channel this suggests where the optimum stays on the boundary. Along the capacity curve the birth lies at $r_m^* = 2.40857, 2.32318, 2.25017, 2.18742, 2.13331$ and 2.08650 for $r_p = 2.5, 2.6, \dots, 3.0$. Below the circle value the center excess is negative wherever we evaluated a converged density, and the solved rings at $r_p = 3$ approach the small negative scalar value as $r_m \rightarrow 0$. We therefore conjecture, from these grids only, that the optimal input stays on the boundary ellipse for every r_m when $r_p \leq 2.4534$, beyond the range $r_p \leq \sqrt{2}$ of [13, Thm. 1], and for larger r_p while $r_m \leq r_m^*(r_p)$. Nothing between $\sqrt{2}$ and 2.4534 was checked on an interior grid, and the thin-ellipse end needs an argument.

C. Past the birth

For capacity the construction continues past the birth. We write the input as $(1 - m_0)$ times a formal density on the

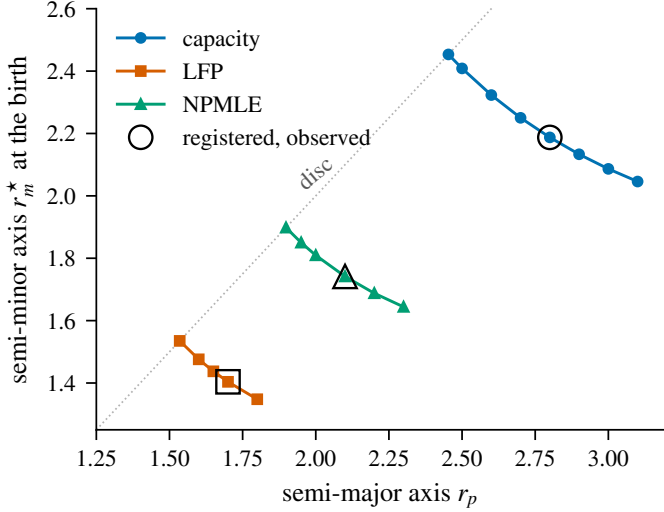


Fig. 3. The formal density predicts where the optimum on a filled ellipse first needs an atom at the center, in all three problems. Lines: predicted $r_m^*(r_p)$ for capacity, the LFP and the NPMLE ($s = 2$); the first point of each curve is the disc. Open symbols: registered births observed on boundary rings (Table V), within 3×10^{-7} of the prediction. Each curve ends where the formal density stops converging.

curve plus m_0 times an atom at the center, and ask that the curve harmonics vanish and the center reach C . At $r_p = 2.8$, $r_m = 2.200$ this predicted the mass $m_0 = 0.0014575498$, and a ring with a center atom gave the same value to 10^{-10} . The Hessian at the center tells when the center atom becomes unstable. On the disc this happens at radius 3.42035, inside the bracket 3.402 to 3.460 that we read from Fig. 5(a) of [27] (a check made after the fact). On the ellipse the major-axis direction goes first: at $r_p = 3.5$ we registered the split at $r_m = 3.20334$, and a 40-atom ring with a center atom changes the sign of that eigenvalue at 3.20334. Past it the center atom becomes a pair on the major axis, in a supercritical pitchfork; the half-separation of the pair, registered at $r_m = 3.22$ as 0.097617, was observed with every printed digit.

D. Rings inside rings

The inner pair is itself a ring, on a curve that nothing prescribes. We describe the inner part by a formal density of its own, of mass m_i , on a free curve $\rho(\theta) = \sum_k \rho_k \cos 2k\theta$. It is fixed together with the boundary density by asking that the optimality function and its radial derivative be flat on the inner curve to the same order. On an ellipse the copying rule is a statement about complex moments, and on the inner curve we read it in the curve's own eccentric parameter u , the angle in $(a \cos u, b \sin u)$ for an inner ellipse of semi-axes a and b . The pair is then lost where the second harmonic of the inner density in u reaches 2, the value of a pair at $u = 0, \pi$ and the largest that a probability law allows; the four-atom ring that follows is lost where the fourth harmonic does. Table VI summarizes the registered tests; the supplementary material lists them in full. The rule passed in all three problems at small inner eccentricity ϵ_i , on ellipses and on the mixed curves. Each problem failed one registered test, at the most elongated inner ring we tried: the capacity pair at $\epsilon_i \approx 0.13$, the LFP pair at 0.048 and the

TABLE VI
REGISTERED LOSSES OF INNER RINGS (THE FULL LIST IS IN THE SUPPLEMENTARY MATERIAL). COUNTS ARE PASSED/REGISTERED; ϵ_i IS THE LARGEST INNER ECCENTRICITY TESTED.

problem	pairs	quartets	failure
capacity	3/4	4/4 ^a	pair at $\epsilon_i \approx 0.13$
LFP	2/3	2/2	pair at $\epsilon_i = 0.048$
NPMLE ($s = 2$)	3/4	2/2	pair at $\epsilon_i = 0.10$

^aone passed only with a fitted second-order band; six further capacity events passed offset bands registered in polar form.

NPMLE pair at 0.10, each lost below the predicted point by 0.4 to 0.9% in the harmonic. The misses grow with ϵ_i , and the capacity quartet passed at $\epsilon_i \approx 0.13$ only because its band included a fitted second-order term. So the inner rings follow the rule to first order in their eccentricity, and the second-order term differs between the problems.

VII. OPTIMAL VALUES IN d DIMENSIONS: THE WIDENED REGION

A. Three value laws

The edge constants of Table I fix the optimal values on regions in two and more dimensions. Write Ω_w for Ω widened by distance w ; for a convex planar region with perimeter P , Steiner's formula gives $|\Omega_w| = |\Omega| + wP + \pi w^2$.

Capacity. On a convex planar region with a smooth boundary whose radius of curvature is large compared with the strip,

$$e^{C+h_2} = |\Omega_{\beta/2}| + \pi(1 + 2k) + c \oint \kappa^2 ds + \dots, \quad (6)$$

where κ is the curvature, $\pi(1 + 2k) = -0.831$, and $c = -0.279$ is fitted on discs and not tested independently. The superellipse test resolves the shape term only to a few per cent (Table VII). For smooth regions whose reach exceeds $\beta/2$ the constant term becomes $(1 + 2k)\pi\chi(\Omega)$, with χ the Euler characteristic, which is zero for an annulus. On the d -ball of radius A , $e^{C+h_d} = V_d A^d + S_d A^{d-1} (\beta/2 + a_d(d-1)/A + \dots)$ with $a_d = \frac{1}{2} + (\beta/2)^2/2 + k = 1.84939$, where V_d and S_d are the volume and surface area of the unit ball. Of these coefficients, β and k come from one dimension, and only c is fitted.

NPMLE. For data uniform on a planar region, $G = DP + D'(0) \oint \kappa ds + b \oint \kappa^2 ds + c_3 \oint \kappa^3 ds + \dots$, where D and $D'(0)$ come from one dimension and b, c_3 are disc fits that depend on the truncation ($b = -0.0180$ or -0.0197 and $c_3 = -0.0803$ or -0.0456 without or with a κ^4 term). This is a perimeter law, and we name no widening width for it. In three dimensions each sphere of a concentric shell carries the ball's surface terms, including the term of order one. With H the mean and K the Gauss curvature, the integrals $\oint H^2 dA$ and $\oint K dA$ both equal 4π on every sphere, so spheres cannot separate them, and other surfaces remain untested. Behind the planar law is an exact identity for the derivative of G under a deformation of the boundary, of which we have a proof sketch.

LFP. By the identity $B = d - I(p_\pi)$, maximizing the Bayes risk is minimizing the Fisher information of the output density $p_\pi = \pi * \varphi_d$ over priors π on Ω . The output density is positive

everywhere, but smoothing does not increase Fisher information (by convexity of $(p, \nabla p) \mapsto |\nabla p|^2/p$), so $I(p_\pi) \leq I(\pi)$. The least Fisher information of a density supported in Ω is $4\lambda_1(\Omega)$, with λ_1 the principal Dirichlet eigenvalue, since $I(\pi) = 4 \int |\nabla \sqrt{\pi}|^2$ and $\sqrt{\pi}$ vanishes on the boundary. Hence $d - B \leq 4\lambda_1(\Omega)$ at the LFP, and for large regions this bound is attained to leading order: this is the second-order term of Levit [30], [31], in which the square of the ground state is the asymptotically least favorable prior [32, notes to Ch. 4]. We find that the next order is a widening by the one-dimensional δ :

$$d - B = 4\lambda_1(\Omega_\delta), \quad (7)$$

on discs, ellipses, annuli (with the coupling of Section VII-D) and balls in $d = 3, 4$ and 5 , up to an excess of the widening that grows with d (Section VII-C).

Known upper bounds on capacity already have the form of a volume plus a boundary term [26], [29]; law (6) replaces their constants by the exact edge strip. Several tests below draw on our earlier computations of the disc and ball NPMLE, of the response of capacity edges to curvature, and of the center events of Section VIII (supplementary material).

B. Tests

Table VII lists the tests. In several of them the optimum had been computed before the prediction was registered and was read only afterwards; the status column marks each row. The LFP solvers covered discs and balls with radius $m \leq 4.5$, ellipses with semi-axes up to 6 and annuli up to 5 wide, where the strip $\delta = 3.35$ is about as wide as the region; they did not converge at $m = 6.5$. At a center event the least favorable prior is a known radial law, with atoms at the center and on a few spheres. The event is the radius at which its optimality function first reaches the constant at a new point. Solving for that radius directly gives exact least favorable priors up to $m = 12.8$ in two dimensions, 13.2 in three, 13.1 in four and 12.1 in five. At the smaller sizes the departures from law (7) are as large as the finite-size departures of the one-dimensional law itself (ratios 1.0139 at $m = 3.5$ and 1.0088 at $m = 4.5$), and a size-matched offset fits them equally well. On ellipses the bands did not separate the widened ellipse from the widened disc of equal area.

C. The LFP strip in two to five dimensions

Read as a widening, the ball's effective strip $2j_{d/2-1,1}/\sqrt{d-B} - m$, with $j_{\nu,1}$ the first zero of the Bessel function J_ν , exceeds the one-dimensional δ by an amount that grows with d (Fig. 4). In two dimensions the excess rises to 0.0148 at $m = 12.8$ (0.44% of δ) and is still rising slowly. In three, four and five dimensions it reaches a maximum, 0.0385 near $m = 9.9$, 0.0635 near 7.9 and 0.0890 near 7.7, and then falls slowly, to 0.0367 at $m = 13.2$, 0.0583 at 13.1 and 0.0817 at 12.1. Whether it returns to δ is not established. Two registered forms for it, a curvature term in $1/m$ and a plateau, failed in $d = 3, 4$ and 5 , and a registered test of its derivative in d at $d = 1$ failed. Between two and three dimensions the excess over the one-dimensional offset

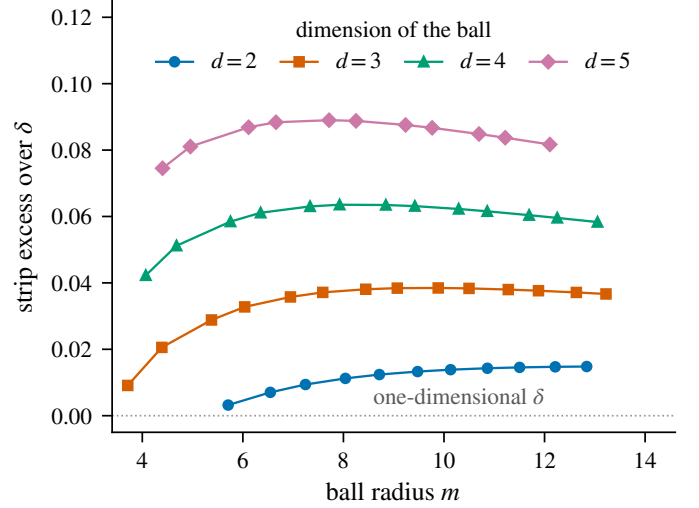


Fig. 4. The LFP strip of a d -ball is wider than the one-dimensional δ , rises to a maximum and then falls slowly; its limit is open. Excess of the effective strip over $\delta = 3.34776$ at the exact center-event states, for $d = 2$ to 5 .

at the same m scales as $d - 1$, to 0.001 at $m = 7.6$ to 8.4 (registered; Table VII).

D. Close walls

When two walls are closer than the edge layers, the local law misses: for the NPMLE on annuli of width $W = 10$ to 18 the registered test failed. The one-dimensional problem explains most of the miss. On an interval of half-width a the edges are already coupled, and the per-end defect of the NPMLE differs from D by an amount $\delta_0(a)$; for capacity, e^{C+h_1} differs from $2a + \beta$ by $\varepsilon(a)$. Taken at half the width and carried along the midline of radius R_{mid} , these deviations predict the misses on annuli.

For the NPMLE the observed miss is 77 to 101% of $4\pi R_{\text{mid}} \delta_0(W/2)$. For capacity the coupling $2\pi R_{\text{mid}} \varepsilon(W/2)$ passes at 19 of 19 entries on narrow annuli, and $4\pi R_{\text{mid}}^2 \varepsilon(W/2)$ at 12 of 14 on a 3D shell; on eccentric annuli the coupling follows the local width. For the LFP the coupling is a factor, the one-dimensional finite-width ratio $\rho_{1D}(W/2)$: annuli narrower than 2δ exceed the plain law by 5 to 7%, and $4\lambda_1(\Omega_\delta)\rho_{1D}(W/2)$ holds on three new annuli to 0.7 to 1.6%. What remains after the coupling, up to 14% (2D) and 33% (3D) of it for capacity and 23% for the NPMLE, has no single sign and is not explained.

VIII. CENTER EVENTS IN d DIMENSIONS

On a d -ball the center alternates between a birth, where an atom appears at the center, and a lift-off, where the center atom becomes a small sphere; on an interval the sequence is birth and split (Section III). To pair the events, count a center atom as one atom and a sphere as two, matching the pair of atoms at $\pm r$ on the interval; the one-dimensional partner of an event on the ball is the interval event with the same count. The first LFP events are the classical thresholds: 1.0567 on the interval [4] and the threshold of the uniform prior on the

TABLE VII

TESTS OF THE VALUE LAWS (RATIO = OBSERVED/PREDICTED). STATUS: R, OPTIMUM COMPUTED AFTER REGISTRATION; E, OPTIMUM EXISTED BEFORE REGISTRATION AND WAS READ AFTER IT; S, INDEPENDENT RE-SOLVE OF A KNOWN STATE; P, READING AFTER THE FACT.

problem and region	result	status	amendment
capacity, disc ($A \geq 10$, 16 states): strip β	2×10^{-5} (three-term fit)	E	1510
capacity, superellipses $p = 2.5$, 3 ($A = 8$)	3.4×10^{-5} , 3.7×10^{-5} in C ; constant term cancels	R	1513–1514
shape term	2.6%, 1.0% (resolution about 6%, 2.3%)	P	1521
capacity, d -balls, $d = 3, 4, 5$	strip = $\beta/2$	E	1516
capacity, curvature coefficient a_d from one dimension	$k = -0.63226$ (registered -0.633 ± 0.03)	E	1520
NPMLE, edge value on disc, annulus, ball	coefficient D , $2D$	S (R at $R = 33.5$)	1482–1489
NPMLE, four ellipses and two $\cos 4t$ stars	within bands; edge-average alternative excluded	R	1505, 1508
NPMLE, term of order one: 3D ball; 1D at a new width	$a_3 = -0.0413180$; width 20 to 5×10^{-7}	ball E and R; width R	1523, 1527
NPMLE, 3D shells	the ball's term of order one is a surface term	R	1541
LFP, discs $m = 2, 2.5, 3.5, 4, 4.5$ (band $[0.85, 1.15]$)	ratios 1.027, 1.017, 1.007, 1.003, 1.002	R	1558, 1564
LFP, ellipses (5, 3.5), (6, 4) (band $[0.97, 1.04]$)	ratios 1.005, 1.003	R	1560
LFP, balls $m = 2.5, 3.5, 4.5$ (band $[0.85, 1.15]$)	ratios 1.006, 0.999, 0.995	R	1566–1567
LFP, narrow annuli with the 1D coupling (band $[0.99, 1.03]$)	ratios 1.016, 1.008, 1.007 (uncoupled 1.073, 1.038, 1.033)	R	1565, 1568
LFP, d -balls at exact event states, $4 \leq m \leq 7.7$, $d = 2$ to 5	ratios 0.999–1.003, 0.993–0.995, 0.987–0.989, 0.981–0.984	E	1584–1585
LFP, widening in $d = 3, 4, 5$ at $9.5 \leq m \leq 13.5$ (two forms)	both forms fail in each d	R	1590, 1593, 1597, 1601–1604
LFP, 3D excess from the 2D and 1D excesses, factor $d-1$ ($m = 7.6-8.4$)	2 of 2 within 0.003 (misses -0.0009)	R	1591, 1602

The LFP bands were set after the one-dimensional values were known; the annulus band after a reading of 1563. The d -ball bands were $[0.98, 1.02]$ for $d \leq 3$ and $[0.95, 1.05]$ for $d \geq 4$.

sphere [28], [35], tabulated in Table I of [7]. The shift Δ_d of each event n from its partner follows, to first order,

$$\Delta_d = (d-1)(u_n + v_n d), \quad (8)$$

with two parameters per event fixed by $d = 2$ and 3. The form was suggested by McMahon's expansion of the zeros of Bessel functions, $j_{\nu,s} \approx (s + \nu/2 - 1/4)\pi - (4\nu^2 - 1)/(8(s + \nu/2 - 1/4)\pi)$ with $\nu = d/2 - 1$, whose two correction terms are linear and quadratic in d . Fitted event by event, however, the two parameters do not behave as one local wavenumber and one radius, so we use (8) as an empirical two-point rule in d .

Registered blind, the rule matched the LFP events e_3 to e_7 in $d = 4$ and 5 within 0.64% and 1.81%, and the NPMLE events in $d = 4$ within 1.31%, 16 of 17 within 1%. On capacity events computed earlier, registered but not blind, it held within 1% in $d = 4$ and 2.8% in $d = 5$. The first LFP events miss in $d = 4$ and 5, by up to 12.7%. In $d = 8$ the first events of every problem miss, by up to 19% for the scored events, and the registered tests there failed. The first LFP event in $d = 8$, known but never scored, misses by 88%: its shift is 2.138 against 1.137 from (8). A simpler form, linear in $d-1$, also stays inside the bands at many later events, with misfits up to ten times those of (8); the early events are where the bands discriminate.

The NPMLE events have a second reading. With uniform data on a ball the NPMLE is a rate-distortion problem, and its center events are the critical temperatures of deterministic annealing for a uniform ball source. The first lies where the variance of the source along an axis equals the noise variance. For a uniform ball of radius A that variance is

$A^2/(d+2)$, so the first event sits at $A = \sqrt{d+2}$. At that radius the critical temperature of deterministic annealing, twice the largest eigenvalue of the source covariance [1], [34], equals the temperature $T = 2$ that corresponds to unit noise variance. The instability of the single center atom there is exact (its second derivative is $A^2/(d+2) - 1$), and the computed first event sits at this radius in $d = 1, 3, 4$ and 8; in the last three we stated it before the run. The blind $d = 4$ test above thus predicted a sequence of annealing transitions from $d = 2$ and 3.

IX. WHAT DOES NOT CARRY OVER

Some structure differs between the problems. Counting a center atom as one ring, the capacity disc ($A = 7$ to 11) and the NPMLE disc ($R = 10$ to 25) have $\lfloor K_{1D}/2 \rfloor$ rings, where K_{1D} is the atom count on $[-A, A]$. This follows from the pairing of events in Section VIII away from the shifted events. The LFP disc has one ring more at $m = 2$ and 3.5, and the registered ring-count test failed at $m = 4$. Its center atoms are born at $m = 1.53499, 3.32183$ and 4.98162 , near the midpoints of consecutive one-dimensional events (1.05674 and 2.00365; 2.89582 and 3.74808; 4.56894 and 5.36404). So the disc gains a ring about half an event spacing early. The midpoint rule missed by 0.0048 (after the fact), 0.00012 (registered) and 0.015 (registered, outside its band of 0.01).

Supports also differ where values agree. On superellipses with $p > 2$, solved at $A = 3$, the capacity wall support breaks into pieces, while the value law holds at $A = 8$. Two registered support readouts failed: the sector counts of the eccentric NPMLE support at two eccentricities, on sampling gaps, and the effective radius read from the LFP output density.

TABLE VIII
REGISTERED TESTS THAT DID NOT PASS

outcome	amendment
failed: form S on the ellipse $r_p = 3$, 18-ring (+1.18%)	1356
failed: extremal form, mixed curve, 18-ring (-2.81%)	1331
failed: inner pairs at the largest ϵ_i (three problems)	1438–1477
failed: NPMLE local law on narrow annuli	1530
failed: tilt explanation of the capacity remainder	1539
failed: NPMLE sector readouts at $e = 2.5$, 3	1548, 1550
failed: plain LFP law on annuli narrower than 2δ	1563
failed: LFP ring count at $m = 4$ (as registered)	1564
failed: LFP bulk shape	1571
failed: 3D first center events, pure radial phase	1573
failed: 2D birth e_5 at the 1D midpoint (0.015, band 0.01)	1574
failed: second lift-offs at the fraction of the first	1577
failed: shared residual, NPMLE and capacity, $d = 8$	1587
failed: LFP center events e_3, e_4 in $d = 8$	1589
failed: LFP widening forms in $d = 3, 4, 5$	1601, 1603, 1604
failed: derivative of the LFP widening in d at $d = 1$	1599
third outcome: $D'(0)$ bias; curved-strip event	1525, 1552
not decidable: NPMLE $R = 45$; $a = 10$; LFP rings	1484, 1529, 1558
solver failed validation: ellipse NPMLE	1491
two tilt constants, not one (NPMLE, capacity)	1503

Open problems. The copying and loss conjectures lack a proof, and the linear-response argument is perturbative. The inner rings follow the rule only to first order, and the second-order term differs between the problems. The limit of the LFP strip in $d \geq 2$ is unknown. The closed-curve rule and the one-dimensional count law are two mechanisms that meet only at the first two scalar changes (Section IV); a single derivation that contains both is not known.

The larger conjecture. The work was guided by a conjecture that this paper does not establish. Whenever a law on a bounded region is chosen optimally and seen through Gaussian noise, its atoms follow one description, with the count, the positions, the edge layers and the births fixed by the same one-dimensional constants. Other problems have this form exactly. A prior that maximizes the information a model's data carry about its parameters is a capacity-achieving input, and Mattingly et al. found the $4/3$ exponent there [10]; we have made no separate test there. A decision made under squared-error loss and a cost on mutual information is a rate-distortion problem [1]. We ran a blind test against twelve published one-dimensional solutions of such problems, from rate-distortion and from rational inattention; the supplementary material lists them with their sources. Our NPMLE solver predicted the number of reproduction points or actions in nine: eight of nine with a uniform source and one of three with a truncated normal one. The next steps are a proof of the count law, the limit of the LFP strip, and registered tests of the planar description in these problems. If the description holds there, an optimal constellation, prior, quantizer or decision rule could be read off a smooth density in each of them. If it fails, the failure will show which part of the mechanism is specific to the three problems treated here.

Table VIII lists the registered tests that did not pass; runs that were not accepted, were stopped or were invalid are listed in the supplementary material.

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APPENDIX A CONSTANTS OF THE CERTIFICATE

Lemma 1. *Let $0 < ar_p < \pi/2$. With one coordinate complexified, $y_j \mapsto y_j + ib$, $|b| < a$, every term of $S(y) = \sum_k w_k \exp(\langle y, x_k \rangle - |x_k|^2/2)$ has argument of modulus less than ar_p , so $\text{Re } S > 0$ and $|\log S(y + ibe_j)| \leq A_0 + B_0(|y_1| + |y_2|)$ with $A_0 = r_p^2/2 + |\log \cos(ar_p)| + ar_p + 2|\sum_k w_k - 1|$ and $B_0 = r_p$. Write $D_\pi(q) = |q|^2/2 + \text{const} - \int \varphi_2(y - q) \log S(y) dy$; the trapezoid sum samples $\log S$ at real grid points only, and a complex q enters only through the Gaussian factor $\varphi_2(y - q)$, so the condition $ar_p < \pi/2$ does not involve q . For complex q with $|\text{Re } q_j| \leq Q_j$ and $|\text{Im } q_j| \leq \sigma$, the trapezoid sum with step h over the infinite grid differs from the integral by at most $4(1 + \eta) e^{(a+\sigma)^2/2 + \sigma^2/2} [A_0 + B_0(Q_1 + Q_2 + 1.6 + 0.484h)] / (e^{2\pi a/h} - 1)$, $\eta = 3e^{-2\pi^2/h^2}$. Truncation at $Y = Nh$ adds at most $e^{\sigma^2} \sum_{(j,l)=(1,2),(2,1)} [(1 + \eta)T_1(Q_j) + B_0T_0(Q_j)((1 + \eta)Q_l + 0.7979 + 0.484h)]$, where $T_0(c) = 2\psi(Y - c)/(Y - c)$, $T_1(c) = 2\psi(Y - c)[(A_0 + B_0c)/(Y - c) + B_0]$ and ψ is the standard normal density (valid for $Y - Q_j \geq 2$).*

Proof sketch. Split the error as $(I - T) \otimes I + T \otimes (I - T)$, so that analyticity is needed in one variable at a time, and apply [22, Thm. 5.1] on each axis, with $|\varphi(u)| = \varphi(\text{Re } u) e^{(\text{Im } u)^2/2}$. The sums against the Gaussian use $\int |u| \varphi(u) du = 0.7979$, $h \sum_n \varphi(nh - c) \leq 1 + 3e^{-2\pi^2/h^2}$ (Poisson summation) and $h \sum_n |nh - c| \varphi(nh - c) \leq 0.7979 + 0.484h$: on each side of its maximum at $|u| = 1$ the function $|u| \varphi(u)$ is monotone, so the sum overcounts the integral by at most one cell of height $\varphi(1) = 0.242$ per side. Tails of decreasing functions are bounded by integrals. $\square$

The same bounds hold for the derivative of $\log S$ in a mass, with $(A_0, B_0) = (1/(m_{\min} \cos ar_p), 0)$, and in an angle, with $(A_0, B_0) = (r_p(a + r_p)/\cos ar_p, r_p/\cos ar_p)$; derivatives along the curve follow by Cauchy estimates on a disc of radius 0.5. For the existence step: $h = 1/28$, $N = 658$ ($Y = 23.5$), $a = 1.45$, 576-bit precision, total bound on D_π of 1.6×10^{-109} . For the curve step: $h = 1/16$, $Y = 18$, 256 bits, the same $a = 1.45$, bounds 4.6×10^{-62} on D_π and 1.4×10^{-60} on $\partial_t^2 D_\pi$. The Taylor remainders along the curve use Cauchy discs of radius $\rho = 1$ in t , on which $|\text{Re } q_j| \leq r_p \cosh 1$ and $|\text{Im } q_j| \leq r_p \sinh 1$; the bound on $|D_\pi - C|$ there comes from the same trapezoid sums with $|\varphi_2(y - q)| = \varphi_2(y - \text{Re } q) e^{|\text{Im } q|^2/2}$.

Existence. The starting point has residual 1.3×10^{-50} in the exact equations; three Newton steps on midpoints (not part of the proof) give $\hat{v}$ with rigorous residual $|F(\hat{v})| \leq 1.96 \times 10^{-108}$. With Y the inverse of the midpoint of the interval Jacobian $J(B)$ over $B = \hat{v} \pm r$, $r_i = \max(10|YF(\hat{v})|_i, 10^{-7} \max_k |YF(\hat{v})|_k)$, the Krawczyk operator $K(B) = \hat{v} - YF(\hat{v}) + (I - YJ(B))(B - \hat{v})$ satisfies $\|I - YJ(B)\|_\infty \leq 8.97 \times 10^{-22}$ and $\max_i |K(B) - \hat{v}|_i / r_i \leq 0.100$, so $K(B) \subset \text{int } B$ and B contains exactly one solution.

Boundary. On each of 32 intervals of half-width 0.02454 the function $t \mapsto D_\pi(z(t)) - C$ is enclosed, uniformly for $v \in B$, by its Taylor polynomial of order 36, a Cauchy remainder on a disc of radius 1, and the quadrature bound. The remainder is 1.07×10^{-58} for the function and 2.37×10^{-52} for its second derivative. Zones $|t - \hat{t}_j| \leq r_j + 0.004$ around the atoms and the Taylor breakpoints give 48 base intervals, 38 free and 10 in atom zones, subdivided adaptively to 230 pieces. The last interval is closed at the ball enclosing $\pi/2$. For every $v \in B$, $D_\pi - C \in [-1.12760361458, -1.12760361456] \times 10^{-50}$ at $(r_p, 0)$.

Interior. $\text{Cov}(X | y) \preceq \max_k |x_k|^2 I$ and $\max_k |x_k|^2 = \cos^2 t_1 + r_m^2 \sin^2 t_1 = 0.99984$, so $\nabla^2 D_\pi \succ 0$; every interior point is a strict convex combination of two boundary points, so $D_\pi < \max_{\partial E} D_\pi \leq C$ there.

REFERENCES

- [1] K. Rose, “Deterministic annealing for clustering, compression, classification, regression, and related optimization problems,” *Proc. IEEE*, vol. 86, no. 11, pp. 2210–2239, 1998.
- [2] J. G. Smith, “The information capacity of amplitude- and variance-constrained scalar Gaussian channels,” *Inf. Control*, vol. 18, no. 3, pp. 203–219, 1971.
- [3] M. N. Ghosh, “Uniform approximation of minimax point estimates,” *Ann. Math. Statist.*, vol. 35, pp. 1031–1047, 1964.
- [4] G. Casella and W. E. Strawderman, “Estimating a bounded normal mean,” *Ann. Statist.*, vol. 9, no. 4, pp. 870–878, 1981.
- [5] Y. Polyanskiy and Y. Wu, “Self-regularizing property of nonparametric maximum likelihood estimator in mixture models,” arXiv:2008.08244, 2020.
- [6] S. Shamaï (Shitz) and I. Bar-David, “The capacity of average and peak-power-limited quadrature Gaussian channels,” *IEEE Trans. Inf. Theory*, vol. 41, no. 4, pp. 1060–1071, 1995.
- [7] A. Dytso, M. Al, H. V. Poor, and S. Shamaï (Shitz), “On the capacity of the peak power constrained vector Gaussian channel: An estimation theoretic perspective,” *IEEE Trans. Inf. Theory*, vol. 65, no. 6, pp. 3907–3921, 2019.
- [8] H. Wang, L. Barletta, and A. Dytso, “An improved lower bound on cardinality of support of the amplitude-constrained AWGN channel,” *IEEE Trans. Inf. Theory*, vol. 72, no. 8, pp. 5392–5399, 2026.
- [9] A. Dytso, S. Yagli, H. V. Poor, and S. Shamaï (Shitz), “The capacity achieving distribution for the amplitude constrained additive Gaussian channel: An upper bound on the number of mass points,” *IEEE Trans. Inf. Theory*, vol. 66, no. 4, pp. 2006–2022, 2020.
- [10] H. H. Mattingly, M. K. Transtrum, M. C. Abbott, and B. B. Machta, “Maximizing the information learned from finite data selects a simple model,” *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 8, pp. 1760–1765, 2018.
- [11] M. C. Abbott and B. B. Machta, “A scaling law from discrete to continuous solutions of channel capacity problems in the low-noise limit,” *J. Stat. Phys.*, vol. 176, no. 1, pp. 214–227, 2019.
- [12] L. Barletta and A. Dytso, “On the evolution of the capacity-achieving input support for the amplitude-constrained AWGN channel,” arXiv:2609.10039, 2026.
- [13] A. Dytso, L. Barletta, and G. Kramer, “On 2×2 MIMO Gaussian channels with a small discrete-time peak-power constraint,” arXiv:2401.17084v1, 2024; also in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2024, pp. 2365–2370.
- [14] T. H. Chan, S. Hranilovic, and F. R. Kschischang, “Capacity-achieving probability measure for conditionally Gaussian channels with bounded inputs,” *IEEE Trans. Inf. Theory*, vol. 51, no. 6, pp. 2073–2088, 2005.
- [15] L. D. Brown, “Admissible estimators, recurrent diffusions, and insoluble boundary value problems,” *Ann. Math. Statist.*, vol. 42, pp. 855–903, 1971.
- [16] C. Hatsell and L. Nolte, “Some geometric properties of the likelihood ratio,” *IEEE Trans. Inf. Theory*, vol. 17, no. 5, pp. 616–618, 1971.
- [17] P. J. Bickel, “Minimax estimation of the mean of a normal distribution when the parameter space is restricted,” *Ann. Statist.*, vol. 9, no. 6, pp. 1301–1309, 1981.
- [18] N. Sharma and S. Shamaï (Shitz), “Transition points in the capacity-achieving distribution for the peak-power limited AWGN and free-space optical intensity channels,” *Probl. Inf. Transm.*, vol. 46, no. 4, pp. 283–299, 2010.
- [19] W. B. Jones, O. Njåstad, and W. J. Thron, “Moment theory, orthogonal polynomials, quadrature, and continued fractions associated with the unit circle,” *Bull. London Math. Soc.*, vol. 21, no. 2, pp. 113–152, 1989.
- [20] B. Simon, *Orthogonal Polynomials on the Unit Circle, Part I: Classical Theory*. Providence, RI: Amer. Math. Soc., 2005.
- [21] F. Johansson, “Arb: efficient arbitrary-precision midpoint-radius interval arithmetic,” *IEEE Trans. Comput.*, vol. 66, no. 8, pp. 1281–1292, 2017.
- [22] L. N. Trefethen and J. A. C. Weideman, “The exponentially convergent trapezoidal rule,” *SIAM Review*, vol. 56, no. 3, pp. 385–458, 2014.
- [23] R. Krawczyk, “Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken,” *Computing*, vol. 4, pp. 187–201, 1969.
- [24] R. E. Moore, “A test for existence of solutions to nonlinear systems,” *SIAM J. Numer. Anal.*, vol. 14, no. 4, pp. 611–615, 1977.
- [25] S. M. Rump, “Verification methods: Rigorous results using floating-point arithmetic,” *Acta Numerica*, vol. 19, pp. 287–449, 2010.
- [26] A. Thangaraj, G. Kramer, and G. Böcherer, “Capacity bounds for discrete-time, amplitude-constrained, additive white Gaussian noise channels,” *IEEE Trans. Inf. Theory*, vol. 63, no. 7, pp. 4172–4182, 2017; arXiv:1511.08742.
- [27] T. Ikeda, M. Hayashi, and T. Tanaka, “Capacity and modulations with peak power constraint,” arXiv:1005.3889v2, 2011.
- [28] J. C. Berry, “Minimax estimation of a bounded normal mean vector,” *J. Multivariate Anal.*, vol. 35, pp. 130–139, 1990.
- [29] A. L. McKellips, “Simple tight bounds on capacity for the peak-limited discrete-time channel,” in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, 2004.
- [30] B. Ya. Levit, “On asymptotic minimax estimates of the second order,” *Theory Probab. Appl.*, vol. 25, no. 3, pp. 552–568, 1980.
- [31] B. Ya. Levit, “Minimax estimation and positive solutions of elliptic equations,” *Theory Probab. Appl.*, vol. 27, no. 3, pp. 563–586, 1982.
- [32] I. M. Johnstone, *Gaussian Estimation: Sequence and Multiresolution Models*, draft, 2011.
- [33] H. Robbins, “An empirical Bayes approach to statistics,” in *Proc. Third Berkeley Symp. Math. Statist. Probab.*, vol. 1, 1956, pp. 157–163.
- [34] K. Rose, E. Gurewitz, and G. C. Fox, “Statistical mechanics and phase transitions in clustering,” *Phys. Rev. Lett.*, vol. 65, no. 8, pp. 945–948, 1990.
- [35] É. Marchand and F. Perron, “On the minimax estimator of a bounded normal mean,” *Statist. Probab. Lett.*, vol. 58, pp. 327–333, 2002.
- [36] B. Rassouli and B. Clerckx, “On the capacity of vector Gaussian channels with bounded inputs,” *IEEE Trans. Inf. Theory*, vol. 62, no. 12, pp. 6884–6903, 2016.
- [37] J. Kiefer and J. Wolfowitz, “Consistency of the maximum likelihood estimator in the presence of infinitely many incidental parameters,” *Ann. Math. Statist.*, vol. 27, no. 4, pp. 887–906, 1956.
- [38] B. G. Lindsay, “The geometry of mixture likelihoods: a general theory,” *Ann. Statist.*, vol. 11, no. 1, pp. 86–94, 1983.